ModelIt!

Student Activity Pack

Growing Meat Without the Animal

Optimizing Cultured Meat Bioreactor Systems from Cell to Steak

Grade 09L3

NGSS HS-LS1-2, HS-ETS1-3

Systems and System Models / Stability and Change

STUDENT NAME _____ DATE _____

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Table of Contents

ModelIt!

1 Pre-Assessment 3

2 Scoring Rubric 4

3 Component Analysis 5

4 Model Recording Page 6

5 Simulation Observations 7

6 Research & Extend 8

7 Reflection Questions 9

8 STEM Challenge 10

1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 What do you think 'lab-grown meat' actually means — how do you grow meat without an animal?

2 Why do you think cultured meat currently costs hundreds of dollars per kilogram while conventional beef costs about $10/kg?

3 Draw a diagram of what you think a bioreactor for growing meat would look like and what conditions it would need to maintain.

4 What factors do you think determine whether cultured meat tastes and feels like real meat?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Cell Growth Rate | Nutrient Medium Concentration | Bioreactor Temperature | Oxygen Supply | Waste Product Accumulation | Scaffold Structure | Texture Quality | Energy Input | Production Cost | Scale Factor

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Lab Prototype

Small-scale, optimized conditions

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Maximum Growth

All inputs maximized for fastest cell division

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Commercial Scale-Up

Large bioreactor, cost-minimized inputs

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Fat Cell Integration — The inclusion of adipocyte (fat) cells alongside muscle cells to create the marbling and flavor profile that makes meat taste rich — requiring different growth conditions from muscle cells

Mechanical Stimulation — The application of cyclic stretching or electrical pulses to cultured muscle cells, mimicking exercise to promote fiber alignment, protein production, and the textural properties of real muscle tissue

Serum-Free Medium Development — The replacement of fetal bovine serum (FBS) — an expensive, animal-derived, and ethically problematic component — with plant-derived or recombinant growth factors, which is essential for both cost reduction and consumer acceptance

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Can we grow a steak in a lab that tastes, feels, and costs the same as one from a cow — and should we? Use evidence from your simulation to support your answers.

1. **How does your model reveal the multi-dimensional optimization challenge of cultured meat production?**

2. **Which parameter do you think is the most critical bottleneck for commercial viability, and what does your model suggest?**

3. **What trade-offs between Production Cost and Texture Quality did you discover, and which compromises are acceptable?**

4. **How would adding Fat Cell Integration change the bioreactor requirements and production complexity?**

5. **What are the environmental implications of cultured meat compared to conventional livestock farming according to your model's Energy Input data?**

8 STEM Challenge: Design a Commercial-Scale Cultured Meat Bioreactor

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YOUR ENGINEERING MISSION

Design a bioreactor system that can produce cultured meat at commercial scale ($15/kg) while maintaining texture quality comparable to conventional beef.

THE SCENARIO

A cellular agriculture startup has proven they can grow cultured beef at lab scale — but it costs $400/kg. They need to reduce costs by 96% while maintaining quality. Your engineering team has been hired to design the bioreactor system that bridges the gap between lab and commercial production.

Engineering Design Thinking

1. WHAT BIOREACTOR CONFIGURATION WOULD YOU USE — STIRRED TANK, HOLLOW FIBER, PERFUSION, OR A NOVEL DESIGN?
2. HOW WOULD YOU MONITOR CELL HEALTH AND PRODUCT QUALITY IN REAL TIME INSIDE THE BIOREACTOR?
3. WHAT RECYCLING OR RECOVERY SYSTEMS WOULD YOU INCLUDE TO REDUCE NUTRIENT MEDIUM WASTE?
4. HOW WOULD YOU INCORPORATE SCAFFOLD STRUCTURES AT COMMERCIAL SCALE?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Bioprocess Engineers design and optimize bioreactor systems for cellular agriculture, biopharmaceutical production, and industrial biotechnology. They work for cultured meat companies (Upside Foods, GOOD Meat, Mosa Meat), biotech firms, and food technology startups, earning $80,000–$150,000/year. Tissue Engineers who specialize in edible scaffold design earn $90,000–$170,000/year.